SIMATS ENGINEERING

ASSESSMENTTOOL1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

COURSECODE: MLA01

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 Time: 60 Min No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty   Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services. As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10

2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	**Weightage (%)**	**Excellent (Full Marks)**	**Good**	**Average**	**Poor**
Problem Understanding &AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.

		performance measures with proper justification.			
Dialogflow Agent Design &Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation &Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation anddecision flow.
Application of AI Techniques &Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.

Testing, Evaluation &Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

Q1. Problem Analysis and Agent Design (10 Marks)

Problem Analysis

Universities receive a large number of student queries every day regarding course registration, timetable, examination schedules, attendance, and campus services. Answering these queries manually takes time and increases the workload of faculty and administrative staff.

Artificial Intelligence (AI) can solve this problem by developing a conversational chatbot that provides instant, accurate, and 24×7 responses. The chatbot uses Natural Language Processing (NLP) to understand user queries, identify their intent, and generate appropriate responses automatically.

Agent Design

Agent Name

StudentSupportAssistant

Objective

To provide instant responses to student queries related to:

- Course Registration
- Timetable
- Attendance
- Examination Schedule
- Campus Services

Users

- Students
- Faculty Members
- Administrative Staff

Environment

The chatbot operates in the university information system and interacts with users through Dialogflow.

Inputs

The chatbot accepts natural language text queries such as:

- "Show my timetable"
- "Register course"
- "What is my attendance?"
- "Exam schedule"
- "Library timing"

Outputs

The chatbot provides appropriate responses such as:

- Registration instructions
- Timetable information
- Attendance information
- Examination schedule
- Campus service details

Performance Measures

The chatbot should:

- Provide accurate responses.
- Respond quickly.
- Understand different forms of user questions.
- Handle unknown queries politely.
- Be available 24×7.

AI Concepts Used

- Natural Language Processing (NLP)
- Intent Recognition
- Entity Recognition
- Context Handling
- Machine Learning

Q2. Dialogflow Agent Development (10 Marks)

Agent Name

StudentSupportAssistant

Intents Created

1. Welcome
2. CourseRegistration
3. Timetable
4. Attendance
5. Examination
6. CampusServices
7. Default Welcome Intent
8. Default Fallback Intent

Entities

course

- AI
- ML
- Python
- Java
- DBMS

service

- Library
- Hostel
- Bus
- Cafeteria

Training Phrases and Responses

Intent	Sample Training Phrase	Response
Welcome	Hello	Hello! Welcome to Student Support Assistant. How can I help you today?
CourseRegistration	Register course	You can register for your courses through the university portal before the registration deadline.
Timetable	Show timetable	Your timetable is available in the student portal under Academic Schedule.
Attendance	Attendance	Your attendance details are available in the ERP portal. Please log in to the student portal to view your attendance percentage.
Examination	Exam schedule	The examination schedule is available on the university examination portal.
CampusServices	Library timing	Campus services information is available on the university website.

Context

Input Context: Student_Query

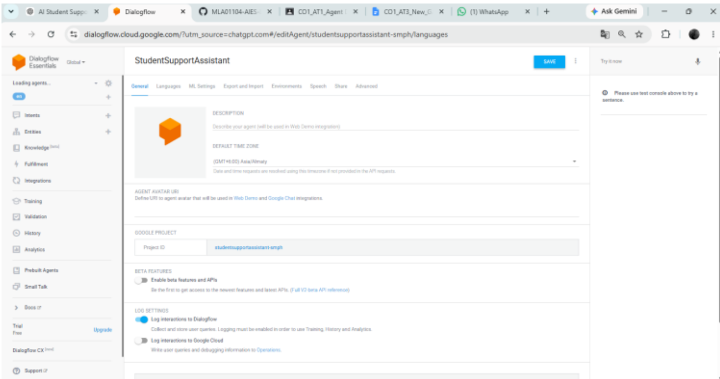
Output Context: Student_Response

Working of the Agent

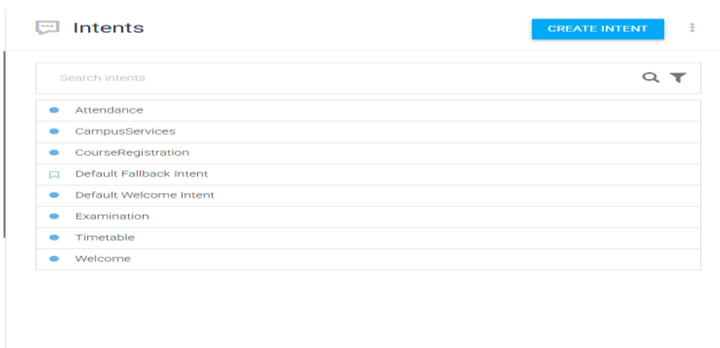
1. User enters a query.
2. Dialogflow analyzes the sentence using NLP.
3. The matching intent is identified.
4. Entities (if any) are extracted.
5. The chatbot generates the predefined response.
6. If no intent matches, the Default Fallback Intent is triggered.

SCREENSHOTS:

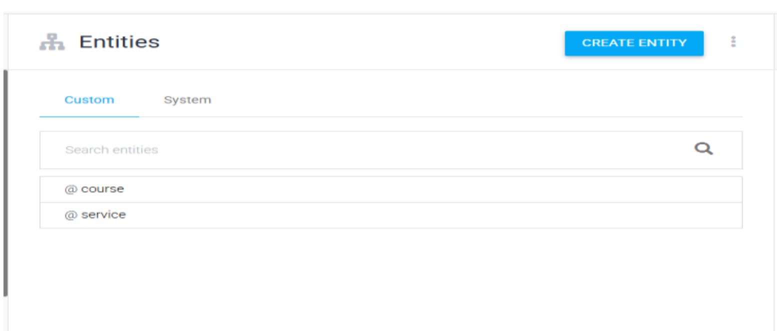
STEP 1: DASH BOARD



STEP 2: INTENTY



STEP 3: ENTINTY



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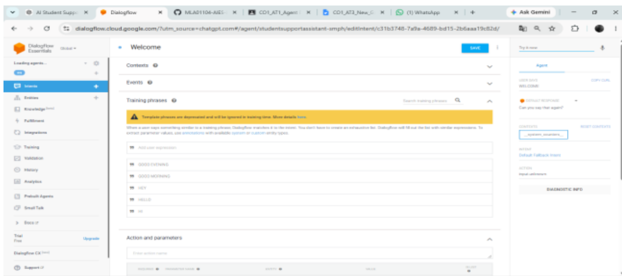
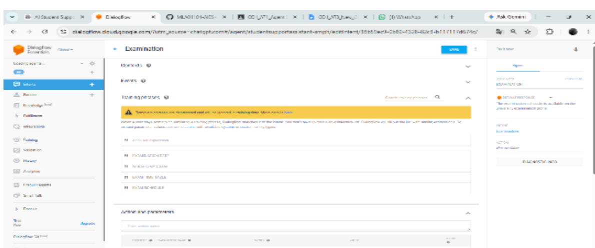
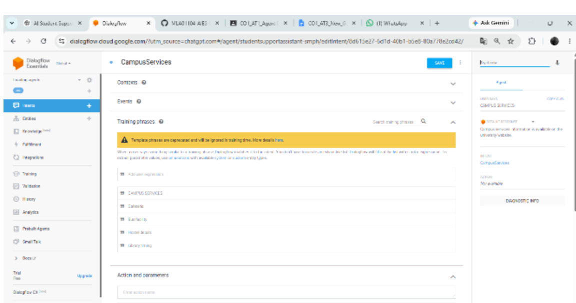
STEP 4:



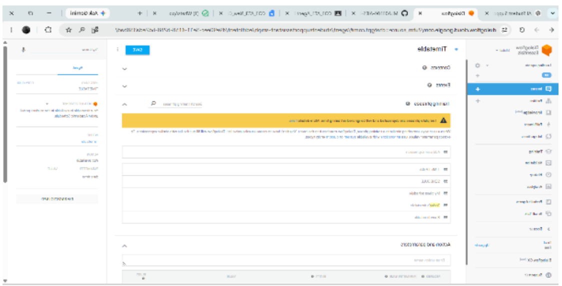
STEP 5:



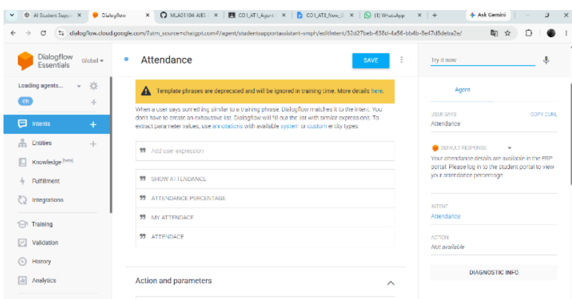
STEP 6:



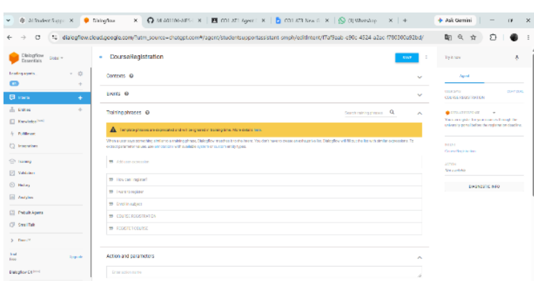
STEP 7:



STEP 8:



STEP 9:



Q3. Problem Formulation and Decision Flow (10 Marks)

Title

Problem Formulation and Conversation Flow Design of StudentSupportAssistant

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1. Initial State

The chatbot is active and waiting for the user to enter a query.

Example:

User opens the StudentSupportAssistant chatbot.

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2. User Goal

The user's goal is to obtain information related to:

- Course Registration
- Timetable
- Attendance
- Examination Schedule
- Campus Services

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3. Possible Actions

The chatbot performs the following actions:

1. Receives the user's query.
2. Analyzes the query using Natural Language Processing (NLP).
3. Identifies the appropriate intent.
4. Extracts entities (if available).
5. Generates the correct response.
6. Asks whether the user needs further assistance.

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4. State Transitions

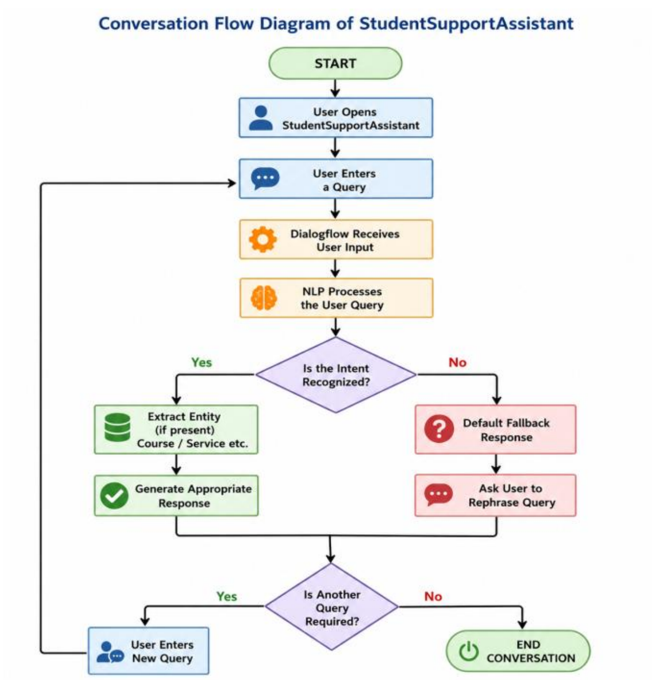
Current State	Action	Next State
Start	User opens chatbot	Waiting for user input
Waiting for input	User enters a query	NLP Processing
NLP Processing	Identify matching intent	Intent Selected
Intent Selected	Generate response	Display Response
Display Response	Ask for another query	Waiting for input or End

5. Goal Completion Condition

The goal is achieved when:

- The chatbot correctly identifies the user's intent.
- The chatbot provides the appropriate response.
- The user's query is answered successfully.

Q3. Conversation Flow Diagram



Q4. **Search Strategy Application** (10 Marks)

Search Technique Used

Breadth First Search (BFS)

Explanation

Breadth First Search (BFS) is an AI search algorithm that explores all possible options at the current level before moving to the next level. In the StudentSupportAssistant chatbot, BFS can be used to model the intent selection process. When a user enters a query, Dialogflow compares the input with all available intents such as Welcome, Course Registration, Timetable, Attendance, Examination, and Campus Services. The chatbot identifies the best matching intent and returns the corresponding response. This approach helps the chatbot quickly recognize user requests and provide accurate answers.

Example

User Query:

"Show my attendance"

Available Intents:

- Welcome
- Attendance
- Course Registration
- Timetable

• Examination

• Campus Services

The chatbot checks the available intents and identifies that the query matches the Attendance intent. It then returns the attendance-related response to the user.

Justification

Breadth First Search is suitable for this chatbot because:

• It systematically explores all available intents.

• It quickly identifies the appropriate intent.

• It improves response accuracy.

• It is simple to implement for small and medium-sized conversational agents.

• It provides fast responses to user queries.

Q5. **Testing and Evaluation (10 Marks)**

Test Cases

Evaluation

Accuracy

The chatbot correctly recognized the user queries and matched them with the appropriate intents. The responses generated were relevant and accurate for the tested scenarios.

Response Quality

• The chatbot responds quickly.

• The responses are simple and easy to understand.

• The interaction is user-friendly and provides useful information.

Unknown Query Handling

When the chatbot receives a query that does not match any predefined intent, the Default Fallback Intent is activated. It politely asks the user to rephrase the question, ensuring that the conversation continues smoothly.

Suggested Improvements

- Integrate the chatbot with the university ERP database for real-time information.
- Provide personalized student details after secure login.
- Support voice-based interaction.
- Add multilingual support.
- Include additional services such as fee payment, results, and placement information.
- Develop a mobile application for easier access.

Conclusion

The StudentSupportAssistant chatbot was successfully developed using Dialogflow. It applies Artificial Intelligence techniques such as Natural Language Processing (NLP), Intent Recognition, and Entity Recognition to understand user queries and provide instant responses related to course registration, timetable, attendance, examination schedules, and campus services. The chatbot reduces the workload of university staff, provides quick and accurate information, and improves the overall student support experience. Q5. Testing and Evaluation (10 Marks)

Test Cases

Test Case	Expected Result	Status
Hello	Greeting displayed	Pass
Register course	Registration response	Pass
Show timetable	Timetable response	Pass
Attendance	Attendance response	Pass
Exam schedule	Examination response	Pass
Library timing	Campus service response	Pass
Random text	Default fallback response	Pass

Evaluation

Accuracy

The chatbot correctly recognized most user queries and produced relevant responses.

Response Quality

- Fast response time

- Easy-to-understand answers
- User-friendly interaction

Unknown Query Handling

When an unsupported query is entered, the chatbot activates the **Default Fallback Intent** and asks the user to rephrase the question.

Suggested Improvements

- Integrate with the university ERP database.
- Provide personalized student information after login.
- Add voice input and voice responses.
- Support multiple languages.
- Develop a mobile application version.

Conclusion

The **StudentSupportAssistant** chatbot was successfully developed using **Dialogflow**. It uses Artificial Intelligence techniques such as **Natural Language Processing (NLP)**, **Intent Recognition**, and **Entity Recognition** to understand user queries and provide instant responses related to course registration, timetable, attendance, examination schedules, and campus services. The chatbot improves response speed, reduces manual workload, and enhances the student support experience.